Math 1200 Lecture 9

Sep 29 2026

Topics: 14.7: Global minima and maxima, optimization Name:_____
Collaborators:_____

Group work 1 For each of the following regions, say whether they are 1) closed and 2) bounded

(a) $D = \{(x, y) \mid x \geq 0, y \geq 0, x^2 + y^2 < 4\}$

(b) $D = \{(x, y) \mid -3 \leq y \leq 3\}$

Group work 2 Find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 - 2x$ on the closed triangle with vertices at $(2, 0), (0, 2)$ and $(0, -2)$.

Group work 3 Find the extreme values of the function $f(x, y) = e^{-xy}$ on the region $D = \{(x, y) \mid x \geq 0, y \geq 0, x^2 + y^2 \leq 4\}$.